OPEN SOURCE / OPEN HARDWARE / POWERED BY APACHE NUTTX

YOUR SETLIST, UNDER YOUR FEET.

Switcher Guitar Plus 2.0

A loop switcher for analog pedals that thinks in songs, not presets.

Every song is split into parts.
Every part remembers its pedals.
Four clip switches play your backing parts over the guitar.



FOURTEEN SWITCHES. ONE SONG AT A TIME.

Seven parts, one bypass, two for the setlist, four for clips. Three knobs for everything else.



Soft-touch momentary switches: no click under your foot, no mechanical latch to wear out.
Clips and AUX play over the guitar and never pass through it: the guitar path stays analog.

ACTIVE PART · READY · EMPTY · BYPASS · CLIP PLAYING

Part switches 1 to 7

Each switch is a part of the current song. Step on CHORUS and every pedal of the chorus comes in at once. Rename parts per song in the app.

Bypass

Takes every loop out of the chain. Step again to return to the part you were playing.

Prev and next song

Walk the setlist in order. The new song opens on its first part.

Clip switches 1 to 4

Tap: plays the clip once.
Hold: the clip keeps looping.
Double tap: stops it.

Three knobs

Push and turn to browse and set things up without the phone.

READ THE SHOW AT A GLANCE.

Song, part and the next move, sized to read from standing height.



3.5-INCH / 480 x 320 / ST7796



COMPACT BUILD / 240 x 320 / ST7789

Setlist and position

Which show, which song, how far in.

Current part

Large enough to read without leaning in.

Next part

What your foot does next.

FX pedals on

Which of the seven pedals are in the chain.

Clips

Green while a clip is playing.

Phone link

Shows when the app is connected.

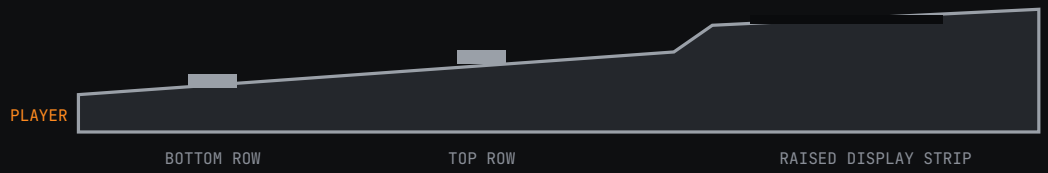
BUILT LIKE A TOOL. OPEN LIKE A LAB.

Every jack at the back. Every switch where your foot expects it.

REAR PANEL



SIDE VIEW



SPECIFICATIONS

Loops	7, true bypass, latching signal relays
Guitar path	100% analog; clips and AUX mixed at the output
Footswitches	14 soft-touch momentary, two rows
Clip player	4 clips, up to 1 minute each, microSD
AUX	Stereo in; to main or to its own outputs
Headphones	2 stereo amps: main mix and AUX
Knobs	3 push-button encoders, out of the foot path
Display	3.5-inch ST7796, or compact ST7789 build
Wireless	BLE for setlists, Wi-Fi for clips
Controller	ESP32-S3 running Apache NuttX
Power	9 V DC, center negative

Firmware, schematics, PCB and enclosure files are published as open source and open hardware.
The Bluetooth and Wi-Fi radio controllers are vendor binaries; everything else is open. Typefaces: OFL. Concept brochure; renderings are illustrations.

THE APP

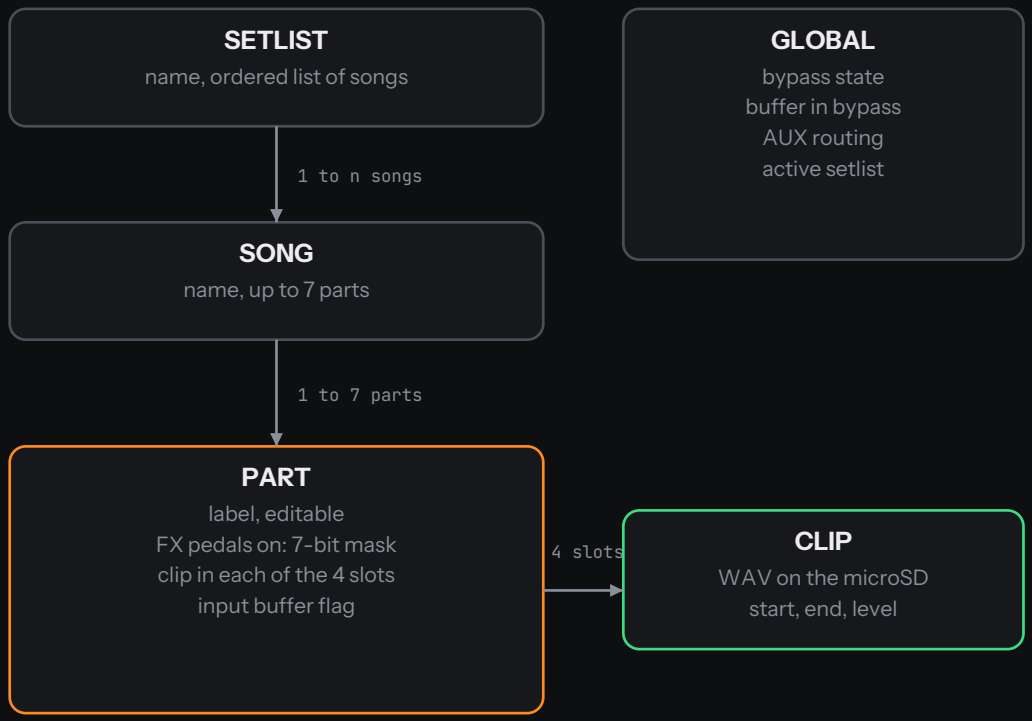


Build the setlist and assign clips at home. The pedal keeps it all. Your phone stays in your pocket.

THE MODEL: SETLIST, SONG, PART.

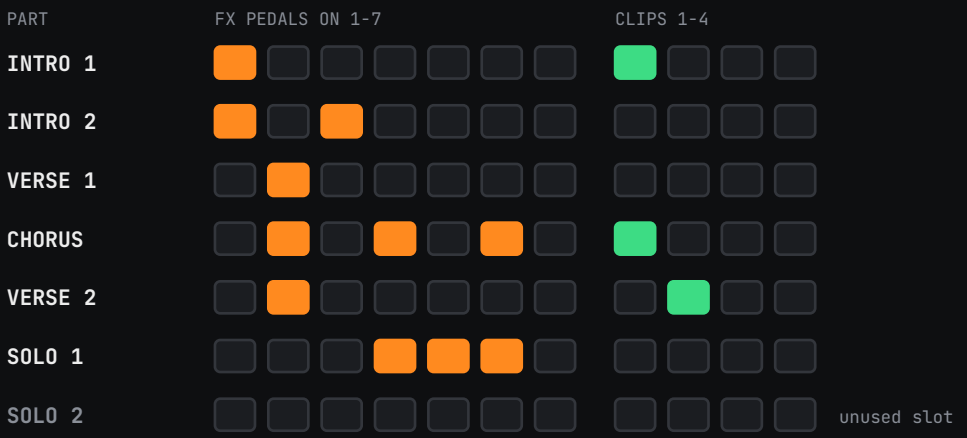
Everything the pedal does on stage is read from this tree.

DECIDED PROPOSED OPEN



DECIDED Bypass, buffer in bypass and AUX routing sit above the songs, in GLOBAL.

EXAMPLE / MIDNIGHT DRIVE



DECIDED Part switch n is bound to part slot n of the current song.

DECIDED An unused slot leaves its switch inactive; the display shows it empty.

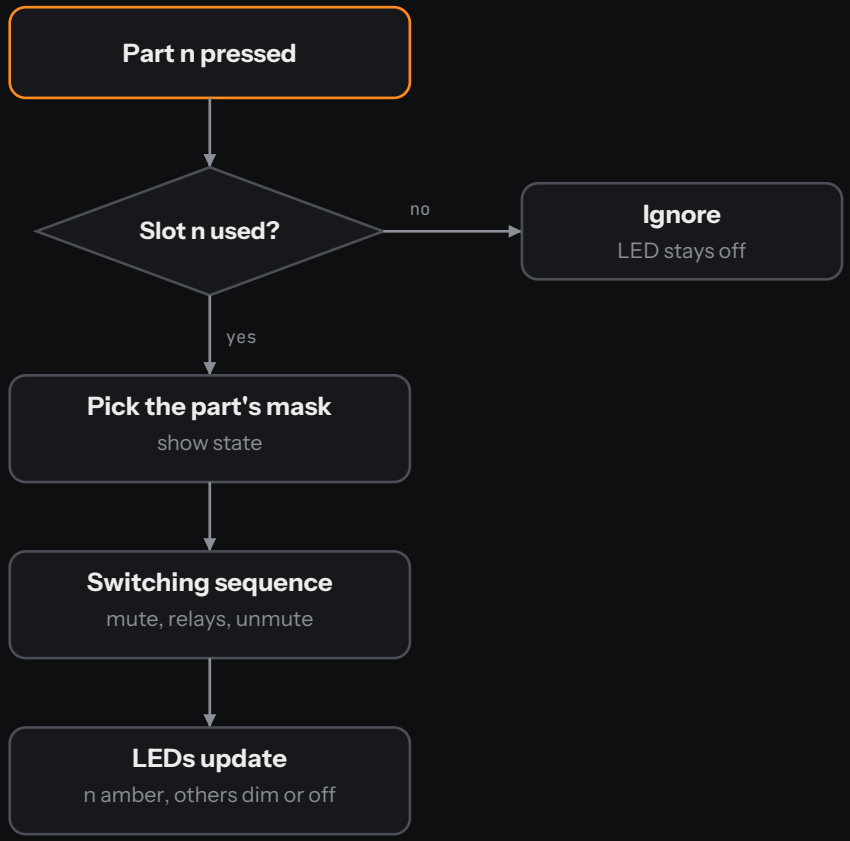
DECIDED Part labels are editable per song; intro 1 to solo 2 are only defaults.

DECIDED Setlists live in flash; clips live on the microSD.

WHAT EACH STEP DOES.

Part, bypass and song switches, from the press to the sound.

DECIDED PROPOSED OPEN



SWITCHING SEQUENCE



PROPOSED Bypass takes every loop out; pressing it again returns to the current part.

DECIDED Input buffer during bypass is a user setting.

PROPOSED Prev and next walk the setlist; a new song opens on its first part.

OPEN What prev does on the first song and next on the last one.

DECIDED At boot every relay is driven to a known state before the mute opens.

DECIDED Latching relays hold their state through a power loss.

CLIPS: TAP, HOLD, DOUBLE TAP.

Four clip switches, read without waiting: the first press always starts the clip.

DECIDED PROPOSED OPEN

GESTURES OVER TIME



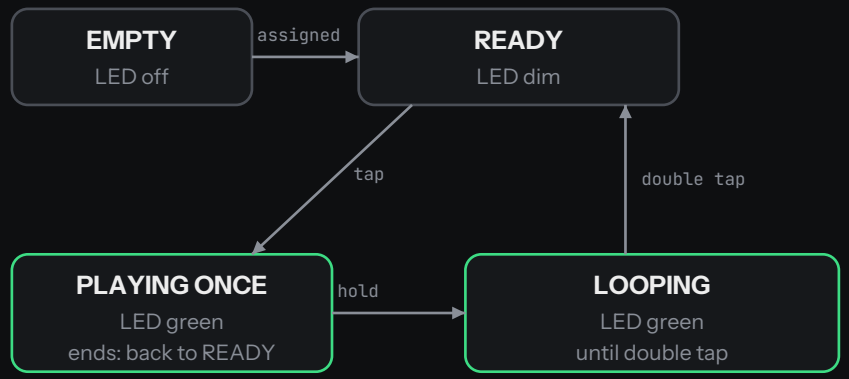
The 2nd press must land within about 300 ms; what played before it is the price of starting instantly. The fade should hide it.

DECIDED 4 slots, up to 1 minute each, mono, read from the microSD.

DECIDED Clips play together and over the guitar; they never pass through it.

DECIDED Each part chooses the clip in each slot.

CLIP SLOT STATES



OPEN Tap on a clip already playing: restart or ignore.

OPEN What playing clips do when the part changes.

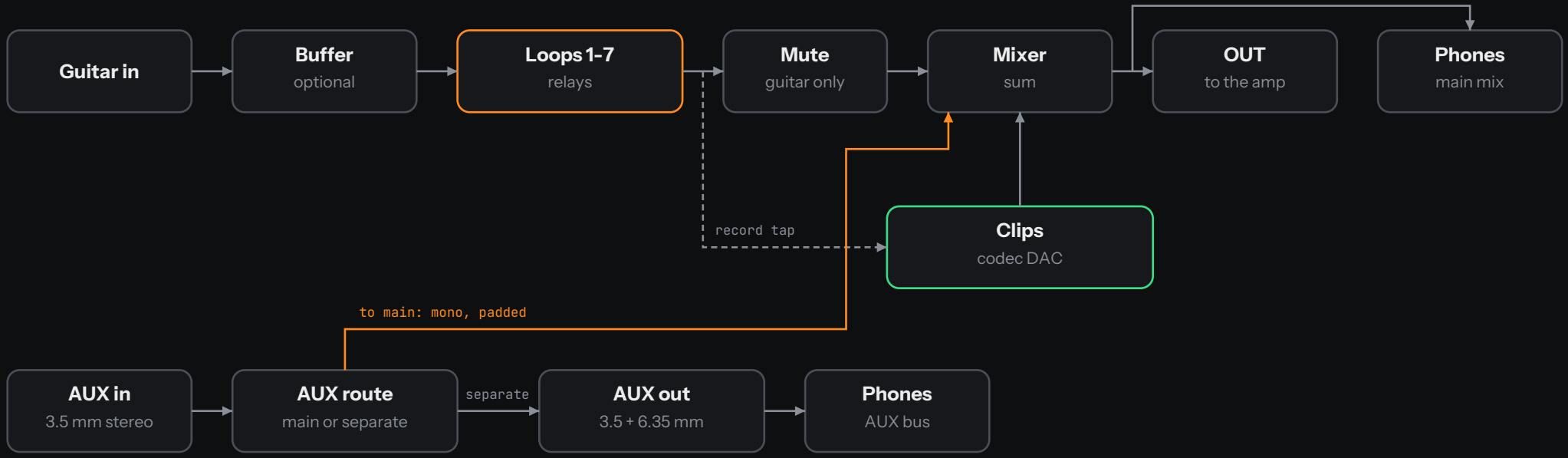
PROPOSED Live recording: a part records into a slot from entering to leaving it; an empty slot records when pressed.

DECIDED Prepared clips: imported and trimmed in the app, synced over Wi-Fi or by card.

WHERE THE SOUND GOES.

One analog guitar path. Clips and AUX join it only at the mixer.

DECIDED PROPOSED OPEN



DECIDED Only AUX IN feeds the AUX bus.

DECIDED To main: AUX is summed to mono and padded down to instrument level.

DECIDED Separate: stereo on 3.5 mm and 6.35 mm jacks in parallel.

DECIDED Two stereo headphone amps; on the main one the guitar sits mono in the centre.

DECIDED The mute covers only the guitar path.

DECIDED Live recording taps the guitar after the loop chain, so it includes the pedals.

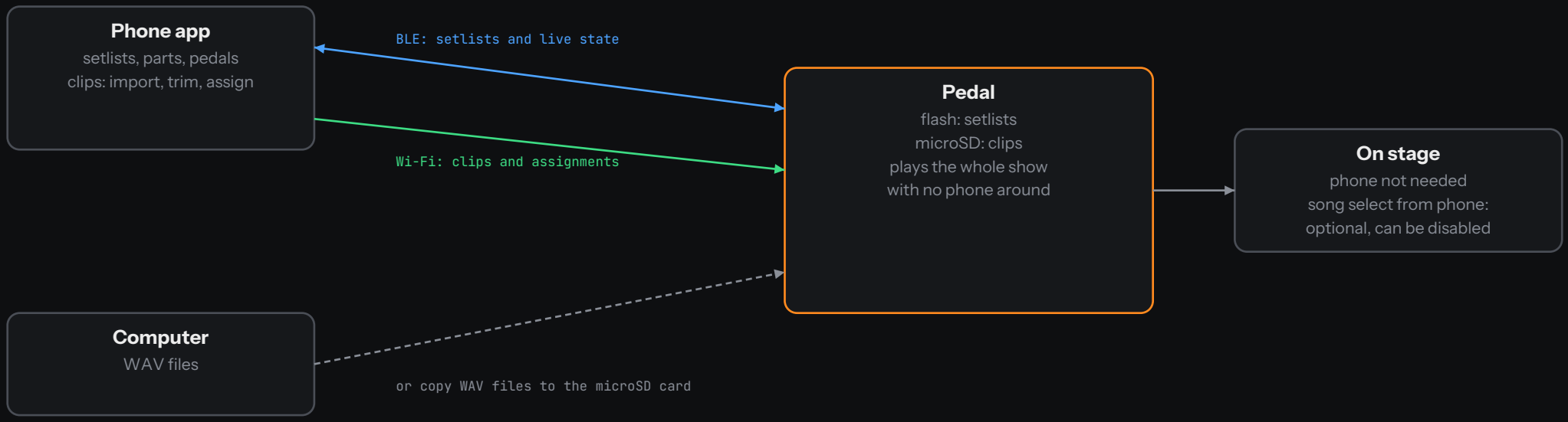
OPEN AUX routing global or per song; headphone volume by pots or by the knobs.

NOTE Guitar on headphones needs an amp and cabinet simulator as the last pedal.

PHONE AT HOME. PEDAL ON STAGE.

The app prepares the show; the pedal plays it alone.

DECIDED PROPOSED OPEN



DECIDED The app organizes before or after a show; it never edits in real time.

DECIDED BLE transfers are chunked; a file is written only after its checksum passes.

DECIDED Pairing uses LE Secure Connections with bonding.

DECIDED Clips never go over BLE: about 5.8 MB per minute would take minutes.

DECIDED The app speaks only the published GATT profile; anyone can write a client.

BACKLOG Pedal order per part: needs a routing matrix and gives up true bypass.

POWER: ONE JACK, FIVE QUIET RAILS.

Noisy loads and audio never share a supply path.

DECIDED PROPOSED OPEN



DECIDED Analog and digital returns are separate and join at one point.

PROPOSED Relay coils run from the raw 9 V with their own return, away from audio ground.

PROPOSED Codec analog supply from a low-noise LDO, never straight from the buck.

PROPOSED Star point at the DC jack; the buck sits far from the audio jacks and op-amps.

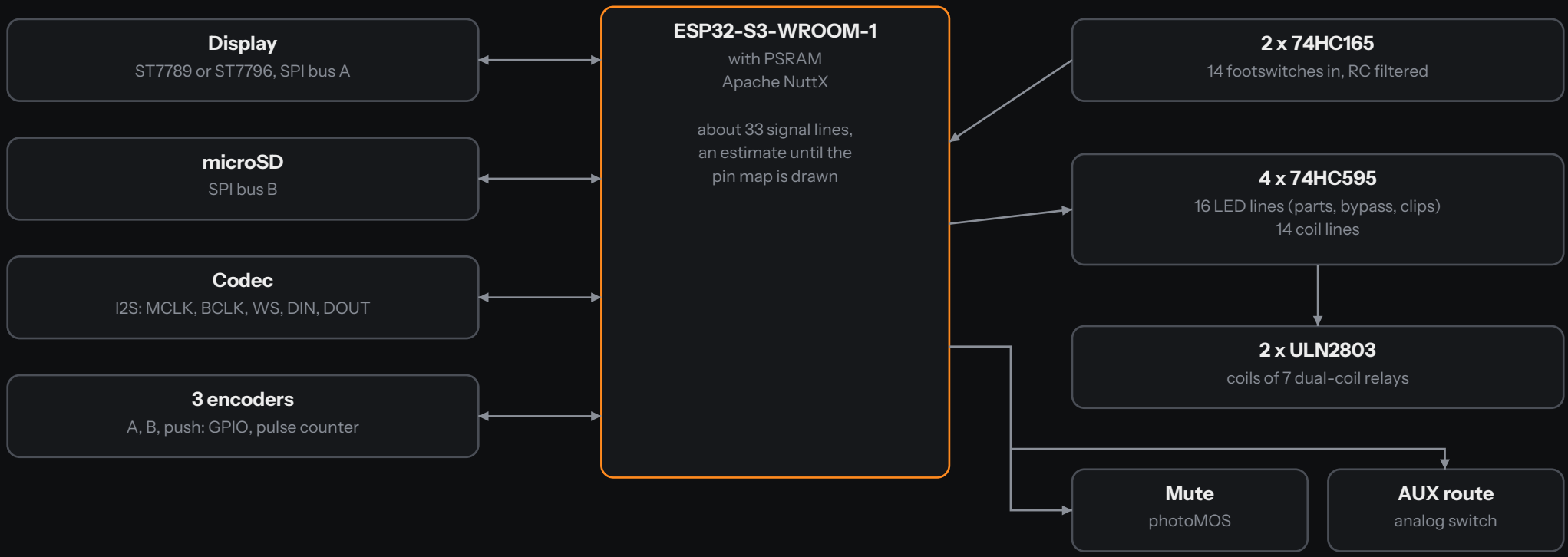
OPEN Adapter rating: needs measured currents (Wi-Fi peaks, backlight, phones, relays).

OPEN Headphone rail topology: single 9 V or a bipolar rail from a charge pump.

CONTROL: ONE CHIP, TWO SHIFT CHAINS.

Every pin of the ESP32-S3 is spoken for, so slow I/O goes through shift registers.

DECIDED PROPOSED OPEN



PROPOSED Display and microSD on separate SPI buses, so a redraw never stalls clip streaming.

PROPOSED Footswitches: RC filter each, read by 74HC165s; firmware debounce on top.

DECIDED Relay coils are pulsed, never held: dual-coil latching relays.

PROPOSED Coils through ULN2803s, whose internal clamp diodes return to the relay rail.

OPEN Pulse width and how many coils fire at once: relay datasheet and supply budget.

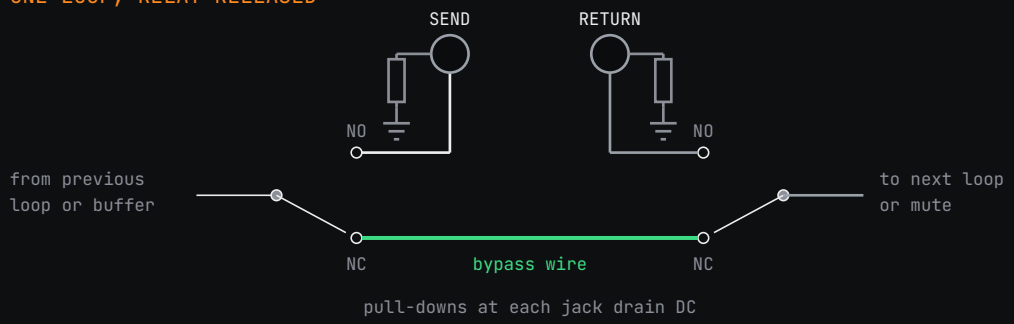
OPEN Programming and debug port, and the final pin map.

AUDIO: RELAYS, ONE MUTE, ONE SUM.

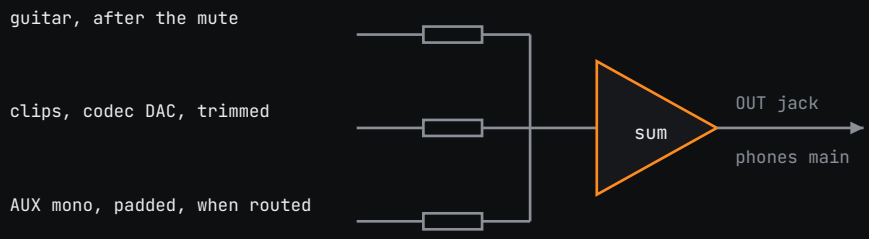
The guitar meets silicon only in the buffer, the mute and the mixer.

DECIDED PROPOSED OPEN

ONE LOOP, RELAY RELEASED



OUTPUT MIXER



STAGE BY STAGE

Input	high-impedance input; OPA1642 unity buffer on 4.5 V ref, bypassable
Loops	7 DPDT latching relays in series, as above
Mute	photoMOS shunts the guitar path to ground while relays move
Record tap	after the loops, scaled to the codec ADC range
AUX	stereo input buffer, route switch, mono sum and pad for main
Phones	two OPA1622-class drivers, one per bus

DECIDED Released relay = loop out; the bypass wire carries the signal.

DECIDED Pull-down on every send and return; 1 Mohm is the starting point.

DECIDED Mute after the loops, before the mixer: clips and AUX never drop out.

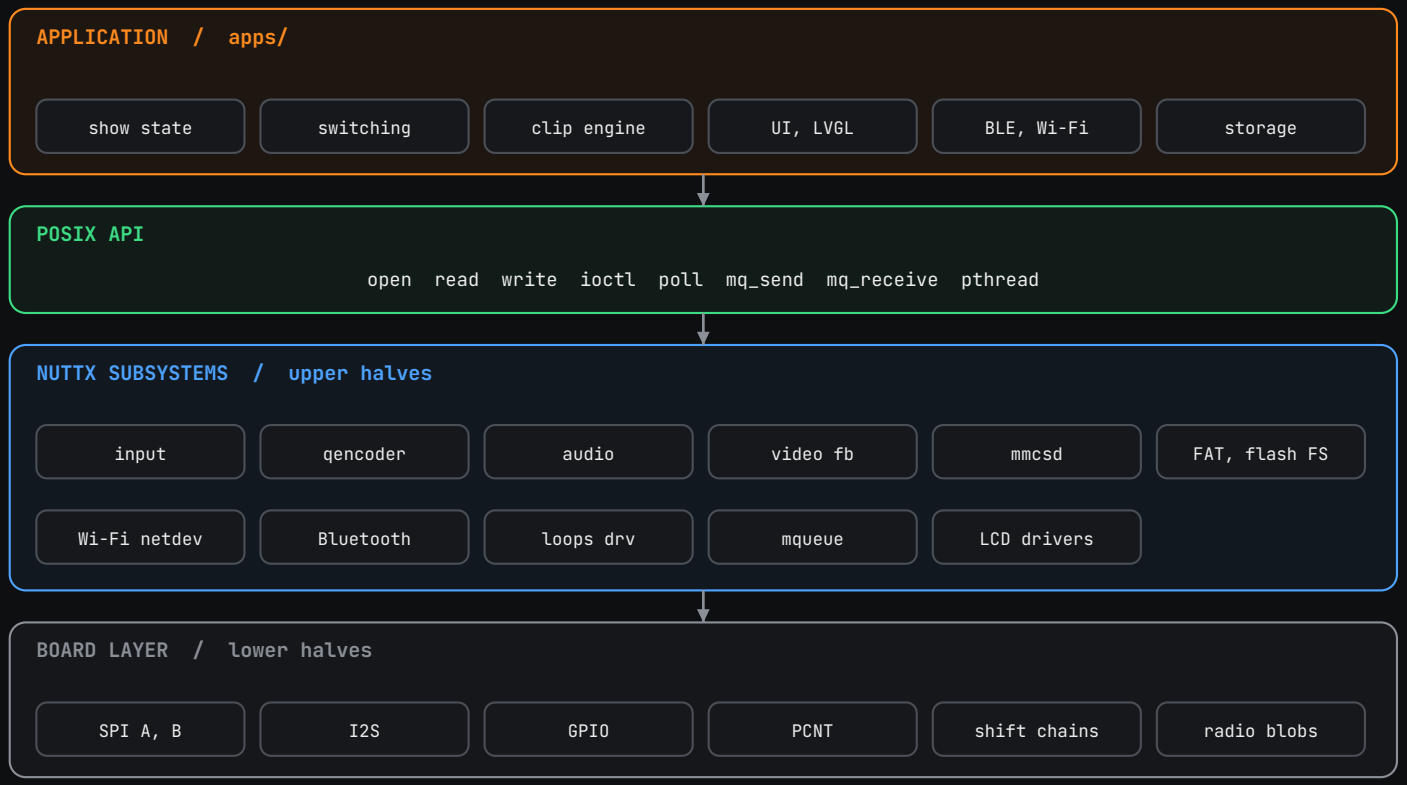
OPEN Mixer topology and gains, codec part, AUX pad value: after level measurements.

NOTE Values here are placeholders until the schematic is drawn and simulated.

FIRMWARE: APACHE NUTTX, POSIX ALL THE WAY.

The application only opens device nodes; the board layer owns the hardware.

DECIDED PROPOSED OPEN



DEVICE NODES

/dev/buttons	14 footswitches
/dev/qe0..2	3 encoders
/dev/audio/pcm0	codec, clips
/dev/fb0	ST7789 or ST7796
/dev/loops	relay mask, mute
/dev/mmcsd0	mounted at /sd
flash partition	mounted at /data
wlan0	clip sync only
Bluetooth host	GATT server

DECIDED The application uses only standard NuttX devices; no register access from apps/.

DECIDED The board layer hides how the display and the relay coils are wired.

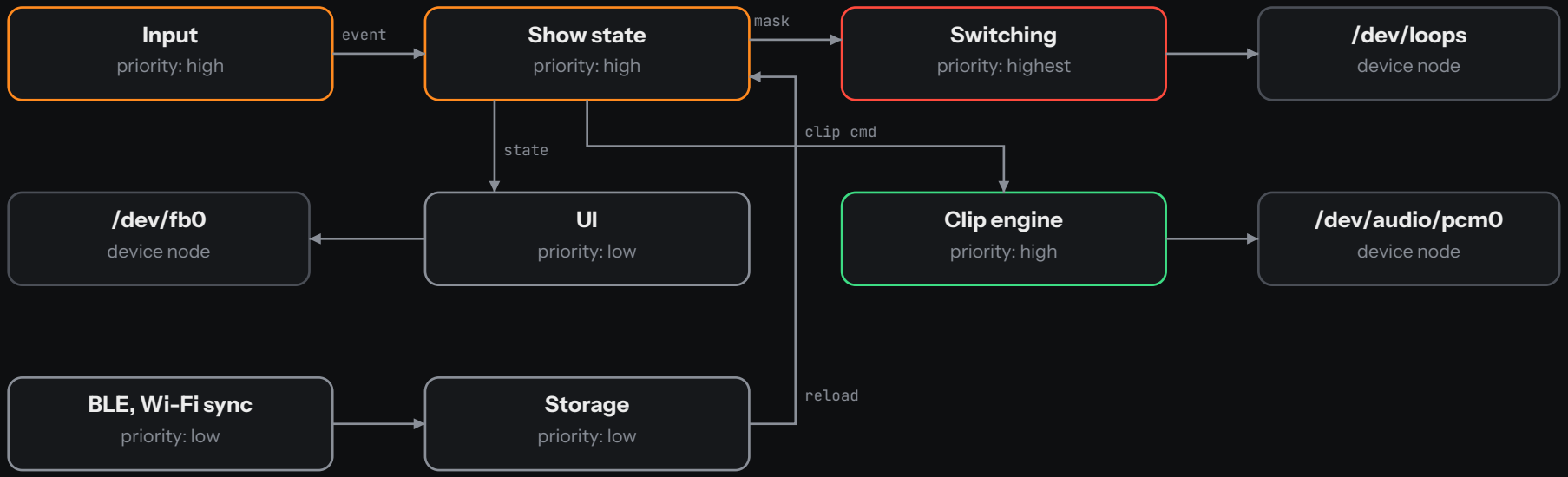
PROPOSED Node names, FAT on /sd, a flash file system on /data, qencoder on the PCNT.

OPEN BLE GATT server on the S3 and the codec driver: check NuttX upstream first.

TASKS, QUEUES, BOOT.

The audio-critical path never waits for a redraw or a radio.

DECIDED PROPOSED OPEN



Every arrow between tasks is a message queue; no task shares state with another.

BOOT

PROPOSED 1. Hardware holds the mute closed.

DECIDED 2. Every relay is driven to a known state.

PROPOSED 3. Mount /data and /sd, load the setlist.

PROPOSED 4. Start UI, input, clips; radios last.

PROPOSED 5. Apply the first part, open the mute.

KCONFIG, IN THE APPLICATION

DECIDED Display: ST7789 or ST7796, sets the UI.

DECIDED Number of loops, default 7.

DECIDED Input buffer present or not.

DECIDED BLE editing on or off.

DECIDED Clip player on or off.